

GENERIC REFINERY - GULF COAST

100 kBPSD Crude & Vacuum Distillation Unit

Preliminary HAZOP Study

Document No.	CFU-000-PR-RPT-002
Revision	A
Status	Issued for review (FEED)
Date	2026-10-02
Prepared	Claude Code
Checked / Approved	- / -

1 Scope and method

This is a preliminary (FEED-stage) HAZOP of the CDU/VDU P&IDs. It uses guideword-deviation analysis (IEC 61882) on 13 nodes. It identifies the main hazards, checks that the proposed safeguards (BPCS, SIS, relief and mechanical) are adequate, and feeds the SIL assessment (LOPA). It does not replace the formal multidisciplinary HAZOP of the issued-for-design P&IDs.

2 Nodes

Node	Description	P&ID	Design intent
N01	Crude charge P-101A/B and cold preheat E-101..E-105	PID-001	130 C, 22 barg
N02	Desalters D-101A/B incl. wash water and brine	PID-002	136 C, 11.5 barg
N03	Hot preheat E-106..E-111 and P-102A/B	PID-003	274 C, 20 barg
N04	Atmospheric heater H-101 incl. fuel gas, APH, fans	PID-004/005	366 C COT
N05	C-101 flash zone, wash zone and stripping section, P-112	PID-006	361 C, 1.5 barg
N06	C-101 pumparounds TPA/MPA/BPA and side strippers C-102/103/104	PID-007/008	150-310 C
N07	C-101 overhead A-101, E-115, D-102, P-103/104/105	PID-009	45-135 C, 0.7-1.2 barg
N08	Stabiliser C-105, E-114, E-116, A-106, D-105, P-115	PID-010	11.5 barg, LPG
N09	Naphtha splitter C-106, E-117, A-107, A-108, D-106	PID-011	0.8 barg
N10	Vacuum heater H-201	PID-012	398 C COT
N11	Vacuum column C-201 incl. wash bed, quench, P-204	PID-013/014	60 mbar(a), 386 C
N12	Ejectors J-201..203, condensers E-202..204, D-201, D-202	PID-015	20-1100 mbar(a)
N13	Fuel gas KO D-103, unit flare KO D-104, steam headers	PID-016	Utilities

3 Worksheet (principal deviations)

Node	Deviation	Causes	Consequences	Safeguards	Recommendation
N01	No flow	P-101 trip, FV-1001 fails closed, tank valve closed	Loss of crude to H-101 passes -> tube overheating / coking, potential tube rupture and fire	FAL-1001; SIF-101 low-low pass flow trips H-101 fuel (SIL 2); P-101B auto-start	Verify SIF-101 response time vs. tube metal heat-up (LOPA)
N01	More pressure	Blocked outlet downstream of P-101	Overpressure of cold train shells	Pump shut-off < design P of cold train (check); PSV-1010 thermal	Confirm shut-off head of P-101 vs. 30 barg exchanger design
N02	Less level (interface)	LV-1007 fails open	Oil carry-under to brine -> WWT upset; low interface shorts grids	LAL-1007; SIF-107 trips desalter transformers; brine oil analyser	Provide brine oil-in-water analyser with alarm
N02	More temperature	TV-1004 bypass fails closed / high crude T	Crude vaporises in desalter, grid arcing, pressure rise	TAH-1004; PIC-1009 backpressure; PSV-1002/1003	Set desalter P >= crude bubble P + 1.5 bar at max T

Node	Deviation	Causes	Consequences	Safeguards	Recommendation
N03	Leak	Tube/flange leak on hot exchangers (> auto-ignition)	Fire in hot train (crude 274 C above AIT)	5Cr / 9Cr materials, fireproofing, deluge on E-111 bank, ROSOV on P-102 suction	Hot flange management programme; flange guards
N04	Less flow (one pass)	Pass FV fails closed, coking	Pass tube overheating and rupture -> firebox fire	FIC-1011..1018 + TDIC-1019; SIF-101 low-low pass flow per pass 2003; TI tube skins	Tube-skin TCs on each pass outlet with alarms
N04	More fuel	PV-1021 fails open, PIC fault	Overfiring, high COT, coking, flame impingement	SIF-103 high-high FG pressure, SIF-106 high-high COT; TAH-1020	
N04	Loss of flame	Low FG pressure, liquid carry-over in FG	Fuel accumulation, firebox explosion on relight	D-103 FG KO with LAHH; SIF-102/104 (BMS per NFPA 85/API 556); purge interlock	BMS to API 556 with 5-volume purge
N04	Less draft	ID fan K-102 trip	Positive firebox pressure, flue gas leakage, burner instability	SIF-105; auto-open stack damper / natural-draft fallback; K-102B	Confirm natural-draft operating capacity (~70 %)
N05	More level	Loss of P-112 / LV-1082 fails closed	Liquid into flash zone -> slugging, tray damage, overflash loss	LAH-1082, SIF-108 (LAHH) stops crude charge	
N05	More water	Desalter upset, water in crude	Rapid vaporisation in column -> pressure surge, tray damage	Desalter interface control; PAH C-101; PSV-1001	Operating procedure for water-slug response
N06	No flow (PA)	P-106/107/108 trip	Loss of heat removal -> column top T and P rise, overhead overload, relief	Spare pumps auto-start; TAH top; PSV-1001 (reflux-failure case)	Include total PA failure in flare load study
N07	Less cooling	A-101 fan failure, CW failure on E-115	Column pressure rise, relief to flare	PAH-1032; 4 fans on 2 buses; PSV-1001 sized for reflux failure	
N07	Corrosion	HCl / NH4Cl salt deposition, low water dew-point margin	Overhead leaks, loss of containment (H2S)	Neutraliser + filming amine (X-103), pH AIC-1037, wash water, Monel top, Ti E-115 tubes, 14 C dew-point margin	Corrosion monitoring: ER probes + chloride analyser on boot water
N08	More pressure	Loss of A-106, reboiler overheat, blocked LPG	Overpressure of C-105 (LPG)	PIC-1091; SIF-110 cuts HP steam; PSV-1005/1006	
N08	Leak	LPG pump seal failure	Flammable cloud, VCE potential	Dual seals Plan 53B, gas detection, ROSOV on D-105 outlet, 30 m spacing to heaters	Consequence modelling of LPG release (QRA input)
N08	Less level	LV/FV on LPG product fails open, LIC-1092 fault	Gas blow-by into LPG treating, overpressure downstream	LALL-1092; SIF-109 closes XV-1093 (SIL 1)	
N09	Less temperature (reboiler)	MP steam loss	Off-spec LN/HN, no safety consequence	TAL-1104; quality alarms	
N10	Less flow / coking	Loss of coil steam, low pass flow	Tube coking, hot spots, rupture	FIC-2007 coil steam, SIF-201	Online spalling / pigging provision
N11	Air ingress	Flange leak under vacuum	Internal fire / explosion in column, loss of vacuum	O2 analyser on off-gas, leak testing, tightness class, hot-oil temperature limits	Add O2 analyser AI-2032 on J-201 suction

Node	Deviation	Causes	Consequences	Safeguards	Recommendation
N11	Less flow (wash oil)	FV-2020 fails closed	Wash-bed coking -> HVGO metals/CCR off-spec, bed plugging	FAL-2020 (critical alarm), min-stop on FV-2020	
N11	More temperature (bottoms)	Loss of quench	VR coking in boot, P-204 damage	TIC-2026 / FIC-2027, TAH-2026	
N12	Loss of vacuum	Motive steam failure, CW failure	Column pressure rise, off-spec, possible relief	PAH-2010, alarm on motive steam P	Vacuum-loss emergency procedure (reduce H-201 firing)
N12	H2S release	Hotwell vent / off-gas line	Toxic exposure	Off-gas routed to H-201 burners with KO D-202; H2S detectors	Backup off-gas route to flare on H-201 trip
N13	Liquid in fuel gas	Upstream carry-over	Burner flame-out / flooding	D-103 LAHH -> SIF-102	

4 Key findings

- The fired heaters dominate the risk profile. SIF-101 and SIF-201 (low-low pass flow, SIL 2) and a BMS to API 556 are essential.
- The hot train and hot pumps handle crude above its auto-ignition temperature. Hot pumps P-102, P-108, P-112 and P-204 get remotely operated shut-off valves (SIF-204), fireproofing to API 2218 and deluge.
- Overhead corrosion (HCl/NH₄Cl) is controlled by desalting, caustic, neutraliser, wash water, Monel and Ti materials, and corrosion monitoring.
- The LPG section is kept at least 30 m from the fired heaters (plot plan), with gas detection and ROSOVs.
- The governing unit relief case (C-101 reflux/pumparound failure, about 220 t/h) needs a global flare-load study covering power and cooling-water failure.